Equations of the IS$_d$ Indicator

Document Singularity Indicator

Blázquez Ochando, M. · Ovalle Perandones, M.A. · Prieto Gutiérrez, J.J.
Universidad Complutense de Madrid

2025

Equation (1) — General structure of the indicator

$$IS_d \;=\; \alpha \cdot O_d \;+\; (1-\alpha)\cdot I_d \qquad \alpha \in [0,1] \tag{1}$$

Where O_d is the textual originality component, I_d the bibliometric impact component, and α the weighting parameter.

Equation (2) — Normalisation of O_d

$$O_d \;=\; \min\left(\frac{O_d^{raw}}{P_{99}(O^{raw})},\; 1\right) \tag{2}$$

Where $P_{99}(O^{raw})$ is the 99th percentile of the O_d^{raw} values in the corpus.

Equation (3) — Raw textual originality O_d^{raw}

$$O_d^{raw} \;=\; \frac{\displaystyle\sum_{i \in \Omega_d} S_i \cdot \log\left(1 + f_i^{doc}\right)}{|\Omega_d| \cdot \log\left(1 + \dfrac{T_d}{|\Omega_d|}\right) + \varepsilon} \tag{3}$$

Where:

- Ω_d = set of unique n-grams present in document d
- $|\Omega_d|$ = number of unique n-grams (cardinality of the set)
- f_i^{doc} = absolute frequency of n-gram i in document d
- T_d = total number of n-grams extracted from document d (with repetition)
- $\varepsilon = 10^{-6}$ = numerical smoothing constant

Equation (4) — Singularity weight S_i

$$S_i \;=\; \underbrace{\log\!\left(\frac{C+1}{cd_i+0.5}\right)}_{\text{Robertson IDF}} \cdot \underbrace{\frac{1}{1+\log(1+f_i^{corpus})}}_{\text{frequency penalty (BM25)}} \tag{4}$$

Where:

- C = total number of documents in the corpus
- cd_i = number of corpus documents containing n-gram i
- f_i^{corpus} = total frequency of n-gram i across the entire corpus

Equation (5) — Normalised bibliometric impact I_d

$$I_d \;=\; \frac{\log(1+C_d)}{\log(1+C_{max})} \tag{5}$$

Where C_d is the number of citations received by document d and C_{max} is the maximum number of citations in the corpus (C_{max} = 53,982 in calcia.db). It holds that $I_d \in [0,1]$ exactly: $I_d = 0$ if $C_d = 0$; $I_d = 1$ if $C_d = C_{max}$.

Equation (α^*) — Objective calibration criterion

$$\alpha^* \;=\; \arg\min_{\alpha\,\in\,[0,1]} |\,\mathrm{skew}(\alpha \cdot O_d + (1-\alpha)\cdot I_d)| \tag{6}$$

On calcia.db: $\alpha^* = 0.692 \approx 0.70$, with $|\mathrm{skew}| = 0.00058$ (global minimum). Kendall's τ of the deciles remains equal to 1.0 for all $\alpha \in [0;\, 0.90]$.